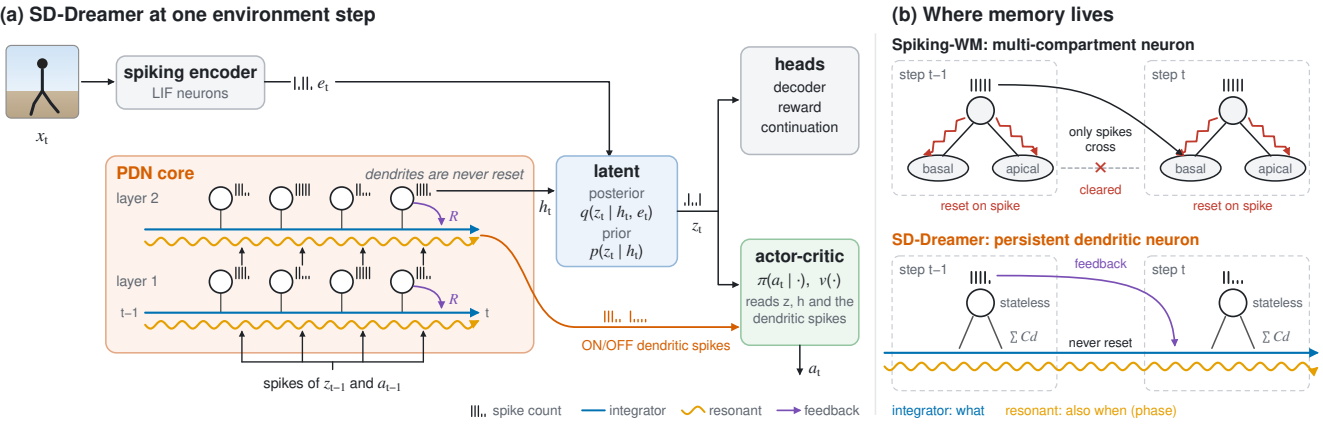

(a) SD-Dreamer at one environment step
x_t
spiking encoder
LIF neurons
e_t
heads
decoder
reward
continuation
PDN core
dendrites are never reset
layer 2
layer 1
t−1
t
R
h_t
latent
posterior
$q(z_t \mid h_t, e_t)$
prior
$p(z_t \mid h_t)$
z_t
actor-critic
$\pi(a_t \mid \cdot),\ v(\cdot)$
reads z, h and the dendritic spikes
a_t
ON/OFF dendritic spikes
spikes of z_{t-1} and a_{t-1}
spike count
integrator
resonant
feedback
(b) Where memory lives
Spiking-WM: multi-compartment neuron
step t−1
step t
basal
apical
only spikes cross
cleared
reset on spike
reset on spike
SD-Dreamer: persistent dendritic neuron
step t−1
step t
feedback
stateless
$\sum Cd$
never reset
integrator: what
resonant: also when (phase)